

Lecture 9

Design for Testability

Prof Peter YK Cheung

Department of Electrical & Electronic Engineering

URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/
E-mail: p.cheung@imperial.ac.uk

In this lecture, we will consider how to modify a digital VLSI design to make it easier to test and verify, especially after it has been fabricated.

Test Approaches

1. Ad-hoc testing
2. Scan-based Test
3. Built-in Self-Test (BIST)

Problem is getting harder

- increasing complexity and heterogeneous combination of modules in system-on-a-chip.
- Advanced packaging and assembly techniques extend problem to the board level

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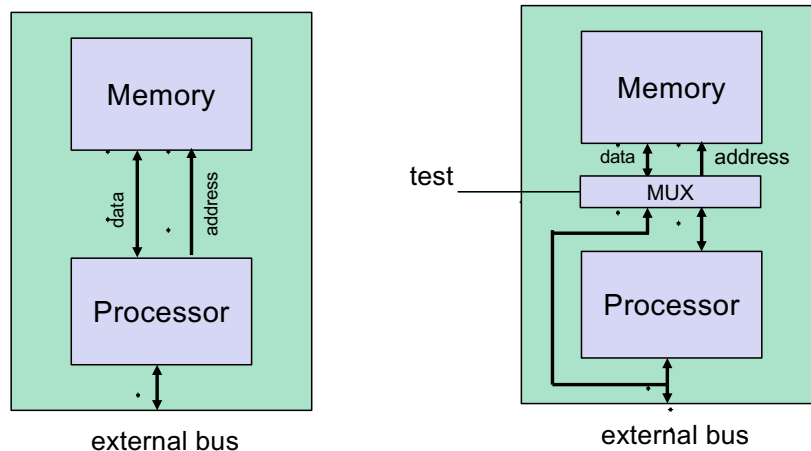
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The keys to designing circuits that are testable are **controllability** and **observability**.

With complexity of VLSI circuits increasing every process generation, it is increasingly important to put testing front and centre in the design process. This approach is known as Design for Testability (DFT).

There are three main approaches to DFT as shown in the slide above.

Ad-hoc Test



Inserting multiplexer improves testability

As suggested by the title ad hoc testing combines a collection of tricks that can be used to increase the observability and controllability of a design that are generally applied in an application dependent fashion.

An example of such a technique is illustrated in the above figure which shows a simple processor with its data memory. Under normal configuration the memory is only accessible through the processor. Writing and reading a data value into and out of a single memory position requires a number of clock cycles. The controllability and observability of the memory can be dramatically improved by adding multiplexers on the data and address buses (right hand side). During normal operation mode these selectors direct the memory ports to the processor. During test the data and address ports are connected directly to the I/O pins and testing the memory can proceed more efficiently. This example illustrates some important DFT concepts.

- a) It is often worthwhile to introduce extra hardware that has no functionality except improving the testability. Designers are often willing to incur a small penalty in area and performance if it makes the design more observable and controllable
- b) DFT often means that extra I/O pins must be provided besides the normal functional I/O pins. The test port in the above figure is such an extra pin. To reduce the number of extra pads that would be required one can mux test signals and functional signals on the same pads. For example the I/O bus in the above figure serves as a data bus during normal operation and provides and collects the test patterns during testing.

An extensive collection of ad hoc test approaches has been devised. Examples include the partitioning of large state machines, addition of extra test points, provision of reset states, and introduction of test buses. While very effective, the applicability of most of these techniques depends upon the application and architecture at hand. Their insertion into a given design requires expert knowledge and is difficult to automate. Structured and automatable approaches are more desirable.

Design for Testability (DFT)

- ◆ Design the chip to increase observability and controllability
- ◆ If each register could be observed and controlled, test problem reduces to testing combinational logic between registers.
- ◆ Better yet, logic blocks could enter test mode where they generate test patterns and report the results automatically.

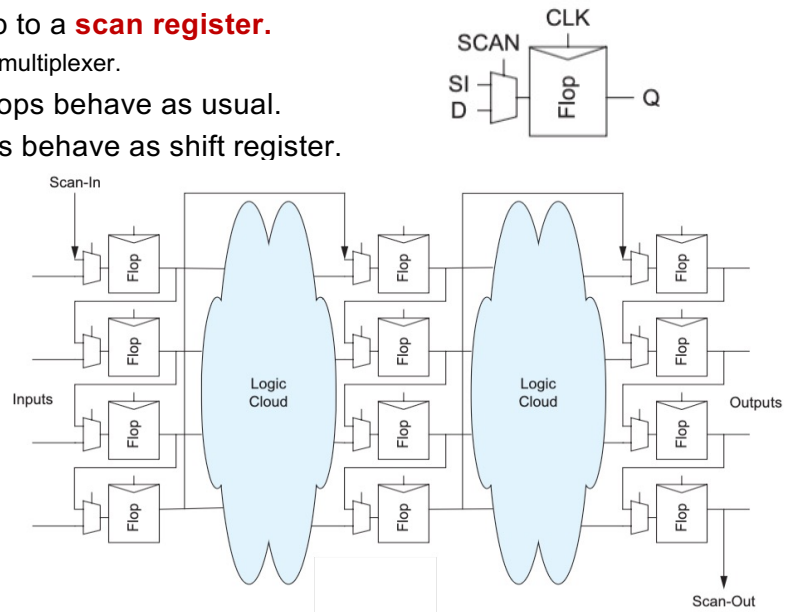
The central idea of DFT is to modify the design to improve both **observability** and **controllability** of the internal nodes. This often involves adding extra hardware to **partition** the circuit into combinational logic blocks between registers which can be tested easily.

To do so, you often need to put the circuit into **test mode**, which is different from **normal mode** when the circuit is under normal operation.

Finally, another approach is to add internal circuits such that **test patterns are generated internally** and the response of the circuit analyzed and reported.

Scan Register

- ◆ Convert each flip-flop to a **scan register**.
 - Only costs one extra multiplexer.
- ◆ **Normal mode**: flip-flops behave as usual.
- ◆ **Scan mode**: flip-flops behave as shift register.
- ◆ Contents of flip-flops can be scanned out and new values scanned in.



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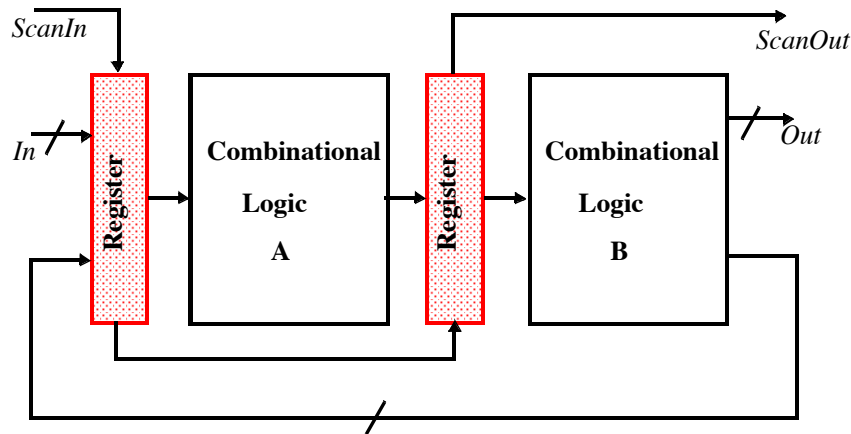
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One way to avoid the sequential test problem is to turn all registers into externally loadable and readable elements. This turns the circuit under test into a combinational entity. These registers are called **scan registers**.

To control a node, an appropriate test vector is constructed, loaded into the registers and propagated through the logic. The result of the excitation propagates to the registers and is latched, after which the contents are transferred to the external world.

Connecting all the registers in a design to a test bus introduces an unacceptable amount of overhead. A more elegant approach is offered by the **serial scan approach** shown above.

Scan-based Test

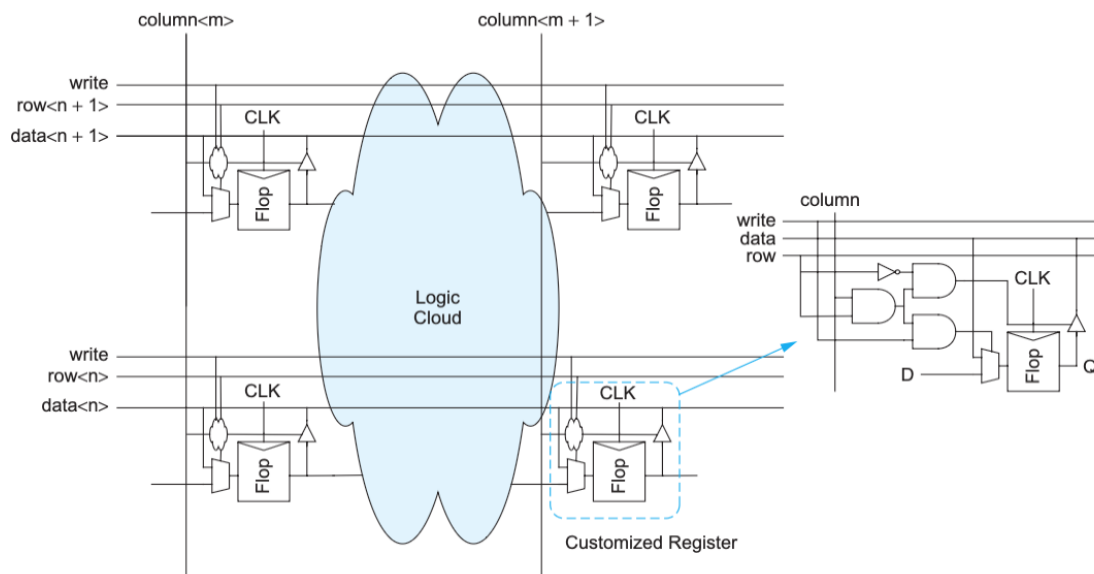


The registers have been modified to support two operation modes. In the **normal mode** they act as N-bit wide clocked registers. During the **test mode** the registers are chained together as a single serial shift register. A test procedure now proceeds as follows:

- 1) An excitation vector for logic module A (and/or B) is entered through pin ScanIn and shifted into the registers under control of a test clock.
- 2) The excitation is applied to the logic and propagates to the output of the logic module. The result is latched into the registers by issuing a single system clock event
- 3) The result is shifted out of the circuit through pin ScanOut and compared with the expected data. A new excitation vector can be entered simultaneously.

This approach incurs only a minimal overhead. The serial nature of the scan chain reduces the overhead.

Parallel Scan Structure



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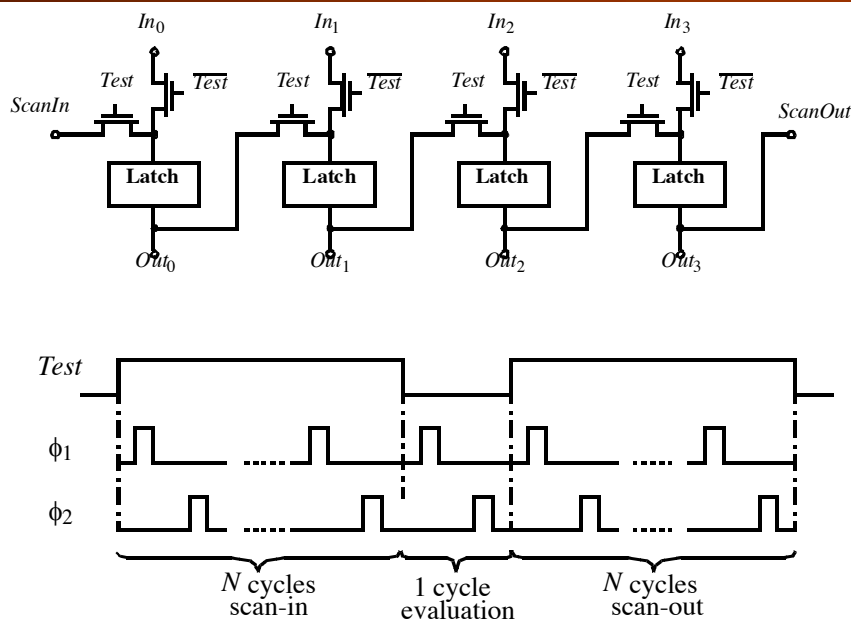
A serial scan circuit can result in a very long chain and slow operation. A better approach is to split the chains into smaller segments as shown above.

The basic idea is to split the logic into segments in rows and columns - similar to how memory arrays are organised. This example shows a two-by-two register section with row and column access signals.

By asserting the row and column access signals in conjunction with the write signal, any register can be read or written in exactly the same method as a conventional RAM.

The circuit on the right shows a possible implementation of such a customized register. In other words, each register in a design is substituted with the shown scan-test flip-flop, making all the internal logic cloud both controllable and observable.

Scan-based Test —Operation



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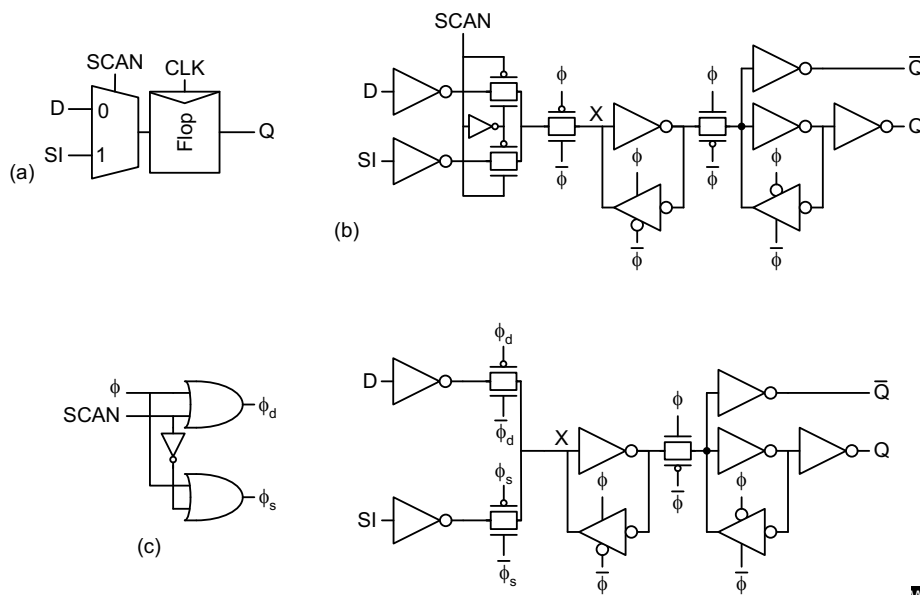
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Traditional registers are easily modified to support the scan technique as demonstrated above (4-bit register extended with a scan chain). The only addition is an extra mux at the input. When test is low, the circuit is in normal operation mode. Setting Test high selects the ScanIn input and connects the registers into the scan chain. The output of the register Out connects to the fan-out logic. It also doubles as the ScanOut pin that connects the ScanIn of the neighboring register. The overhead in both area and performance is small and can be limited to less than 5%.

The bottom figure depicts the timing sequence that would be employed for the circuit under the assumption of a two-phase clocking approach. For a scan chain N registers deep, the Test signal is raised, and N clock pulses are issued, loading the registers. Test is lowered and a single clock sequence is issued, latching the results from the combinational logic into the registers under normal circuit operation conditions. Finally, N extra pulses with $Test = 1$ transfer the obtained result to the output.

Note again that the scan out can overlap with the entering of the next vector.

Scannable Flip-flops



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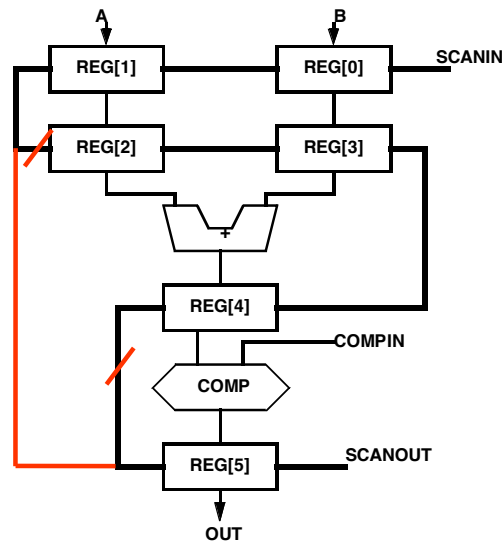
An ordinary flip flop can be made scannable by adding a mux on the data input (Fig. (a)).

Figure (b) shows a circuit design for such a scan register using a transmission gate mux. The setup time increases by the delay of the extra transmission gate in series with the D input as compared to the ordinary static flip-flop.

Figure (c) shows a circuit using clock gating to obtain nearly the same set up time as the ordinary flip flop. In either design, if a clock enable is used to stop the clock to unused portions of the chip, care must be taken that phi always toggles during scan mode.

During scan mode the flip flops are connected back-to-back. Clock skew can lead to hold time problems in the scan chain. These problems can be overcome by adding delay buffers on the SI input to flip flops that might see large clock skews. Another approach is to use non overlapping clocks to ensure hold times.

Scan-Path Testing



Partial-Scan can be more effective for pipelined datapaths

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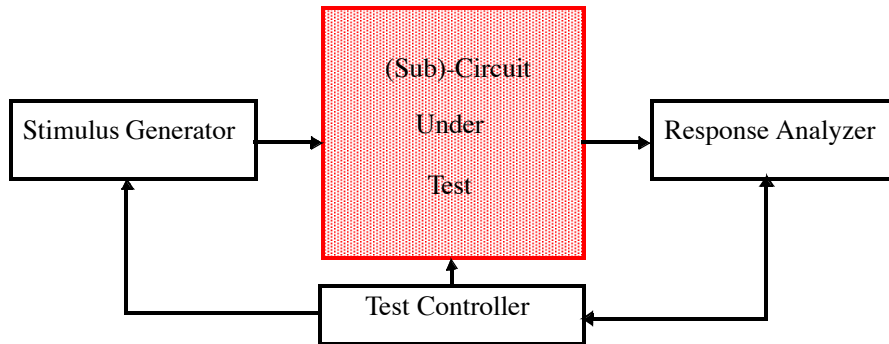
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It is not always necessary to make all the registers in the design scannable. Consider the pipelined datapath above. The pipeline registers in this design are only present for performance reasons and do not strictly add to the state of the circuit. It is therefore meaningful to make only the input and output registers scannable. During test generation the adder and the comparator can be considered together as a single combinational block. The only difference is that during the test execution two cycles of the clocks are needed to propagate the effects of an excitation vector to the output register.

This approach is called partial scan and is often employed when performance is of prime interest. The disadvantage is that deciding which registers to make scannable is not always obvious and may require interaction with the designer.

Built-in Self-test

- ◆ Built-in self-test lets blocks test themselves
 - Generate pseudo-random inputs to comb. logic
 - Combine outputs into a *syndrome*
 - With high probability, block is fault-free if it produces the expected syndrome



Rapidly becoming more important with increasing chip-complexity and larger modules

The general format of a built-in self-test design is shown above. It contains a means for supplying test patterns to the device under test and a means of comparing the device's response to a known correct sequence.

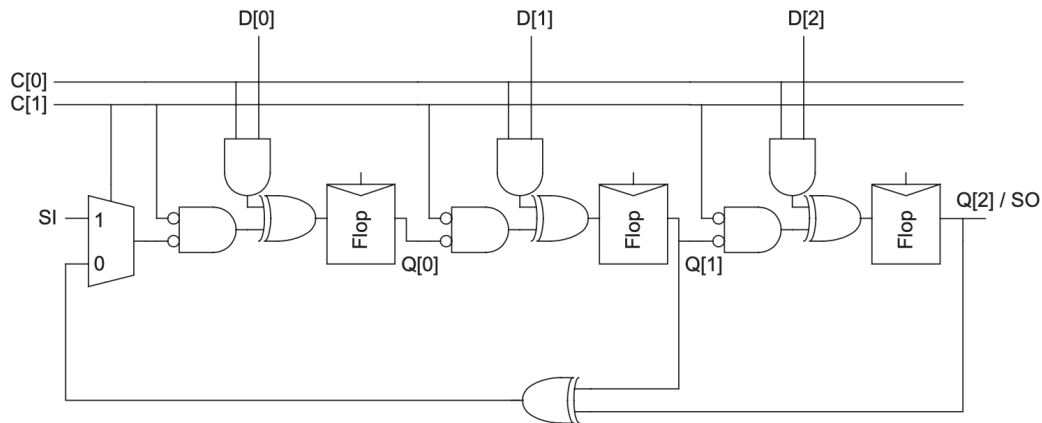
Depending on the nature of the circuit this might require the addition of extra circuitry for the generation and analysis of the patterns. Some of this hardware might already be available as part of the normal operation and the size overhead of the self-test can be small.

BILBO circuit

◆ Built-in Logic Block Observer (**BILBO**)

- Combine scan with PRSG & signature analysis

MODE	C[1]	C[0]
Scan	0	0
Test	0	1
Reset	1	0
Normal	1	1



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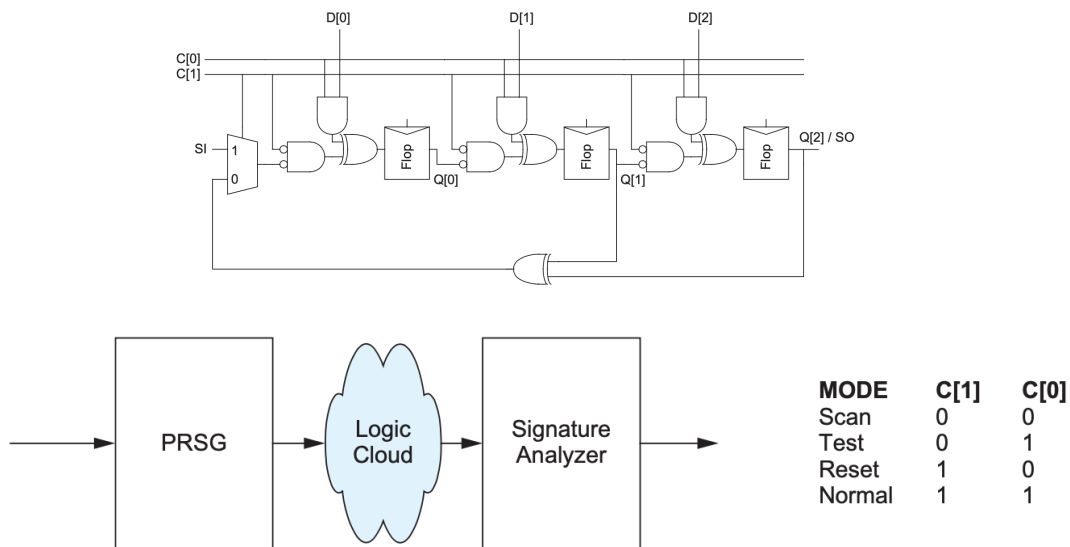
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BILBO represents a modification of the linear feedback shift register and has the advantage that the same hardware can be used for both pattern generation and signature analysis. Each incoming data word is successively XOR'd with the contents of LFSR. At the end of the test sequence the LFSR contains the signature or syndrome of the data sequence which can be compared with the syndrome of the correct circuit. The circuit not only implements a random pattern generator and signature analyzer but also can be used as normal register and scan register depending on the values of the control signals B0 and B1. This test approach that combines all different techniques is known as built in logic block observation (BILBO).

The combination of the signature analysis and the scan technique creates Built In Logic Block Observation (BILBO). The 3-bit BILBO register shown above is a scannable, resettable register that can also serve as a pattern generator and signature analyzer. C[1:0] specifies the mode of operation. In the reset mode of operation.

BILBO in Action



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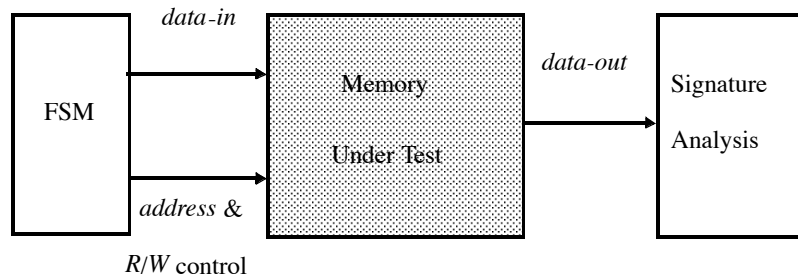
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In the reset mode (10), all the flip flops are synchronously initialized to 0. In normal mode (11), the flip flops behave normally with their D input and Q output. In scan mode (00) the flip flops are configured as 3-bit shift register between SI and SO. Note that there is an inversion between each stage. In test mode (01) the register behaves as pseudorandom sequence generator or signature analyzer. If all the D inputs are held low the Q outputs loop through a pseudorandom bit sequence which can serve as the input to the combinational logic. If the D inputs are taken from the combinational logic output they are swizzled with the existing state to produce the syndrome. In summary, BILBO (BIST ? - Weste) is performed by first resetting the syndrome in the output register. Then both registers are placed in the test mode to produce the pseudorandom inputs and calculate the syndrome. Finally the syndrome is shifted out through the scan chain.

Various companies have commercial design aid packages that support BIST.

Memory Self-Test



Common data test patterns

Pattern Name	Description	Purpose
All-Zeros (0x00)	Write all 0s, then read back	Detect stuck-at-1 faults
All-Ones (0xFF)	Write all 1s, then read back	Detect stuck-at-0 faults
Checkerboard (0xAA / 0x55)	Alternating 1s and 0s: 10101010 / 01010101	Detect coupling faults between adjacent cells
Walking 1 / Walking 0	Move a single '1' (or '0') across all bit positions	Detect bitline and wordline coupling, address decoder faults
Galloping	Move a '1' through memory, writing and reading surrounding cells	Detects cell-to-cell interference (coupling faults)
Random Data Pattern	Pseudorandom or LFSR-generated data	Detects pattern-sensitive faults

Self test is extremely beneficial when testing regular structures such as memories. It is not easy to ensure that a memory which is a sequential circuit is fault free. The task is complicated by the fact that the data value read from or written into a cell can be influenced by the values stored in the neighboring cells because of cross coupling and other parasitic effects.

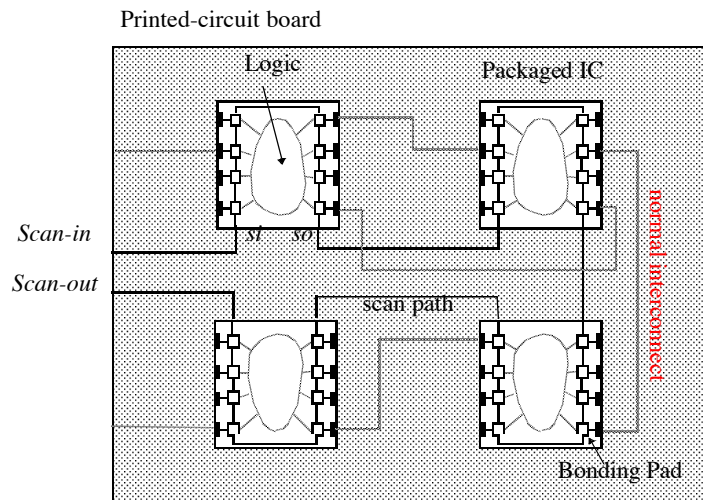
Memory tests therefore include the reading and writing of a number of different patterns into and from the memory using alternating addressing sequences. Typical patterns can be all zeros or ones, or checkerboards of zeros and ones. Addressing schemes can include the writing of the complete memory, followed by a complete read out or various alternating read write sequences. With a minimal overhead compared with the size of a memory this test approach can be built into the IC itself. This approach significantly improves the testing time and minimizes the external control. Applying self test is bound to become more important with the increasing complexity of integrated components and the growing popularity of embedded memories.

Board level testing – JTAG Boundary Scan

- ◆ Testing boards is also difficult
 - Need to verify solder joints are good
 - Drive a pin to 0, then to 1
 - Check that all connected pins get the values
- ◆ Through-hole boards used “bed of nails”
- ◆ SMT and BGA boards cannot easily contact pins
- ◆ Build capability of observing and controlling pins into each chip to make board test easier

Until recently the test problem was most compelling at the integrated circuit level. Testing circuit boards was facilitated by the abundant availability of test points. The through-hole mounting approach made every pin of a package observable at the back side of the board. For test it was sufficient to lower the board onto a set of test probes (bed of nails) and apply and observe the signals of interest. The picture changed with the introduction of advanced packaging techniques such as surface mount or multichip modules. Controllability and observability are not readily available anymore, because the number of probe points is dramatically reduced. This problem can be addressed by extending the scan-based test approach to the component and board levels.

JTAG Boundary Scan

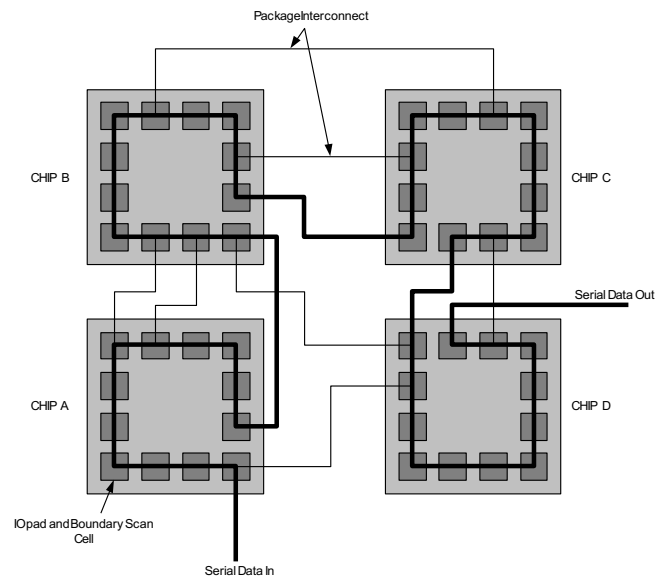


Board testing becomes as problematic as chip testing

The resulting approach is called boundary scan and has been standardized to ensure compatibility between different vendors (IEEE 1149). In essence, it connects the input – output pins of the components on a board into a serial scan chain as shown above. During normal operation the boundary scan pads act as normal input output devices. In test mode vectors can be scanned in and out of the pads providing controllability and observability at the boundary of the components (hence the name). The test operation proceeds along similar lines as described in the previous paragraph. Various control modes allow for testing the individual components as well as the board interconnect. The overhead incurred includes slightly more complex input output pads and an extra on chip test controller (FSM with 16 states). Boundary scan is now provided in most commodity components.

Boundary Scan was originally developed by the Joint Test Access Group and hence it is commonly referred to as JTAG.

JTAG Boundary Scan Example



Boundary scan architecture is shown in this figure. All of the I/O pins of each IC on the board are connected serially in a standardized scan chain accessed through the Test Access Port (TAP) so that every pin can be observed and controlled remotely through the scan chain. At the board level, ICs obeying the standard can be connected in series to form a scan chain spanning the entire board. Connections between ICs are tested by scanning values into the outputs of each chip and checking that those values are received at the inputs of the chips they drive. Moreover chips with internal scan chains and BIST can access those features through boundary scan to provide a unified testing framework.

JTAG interface signals

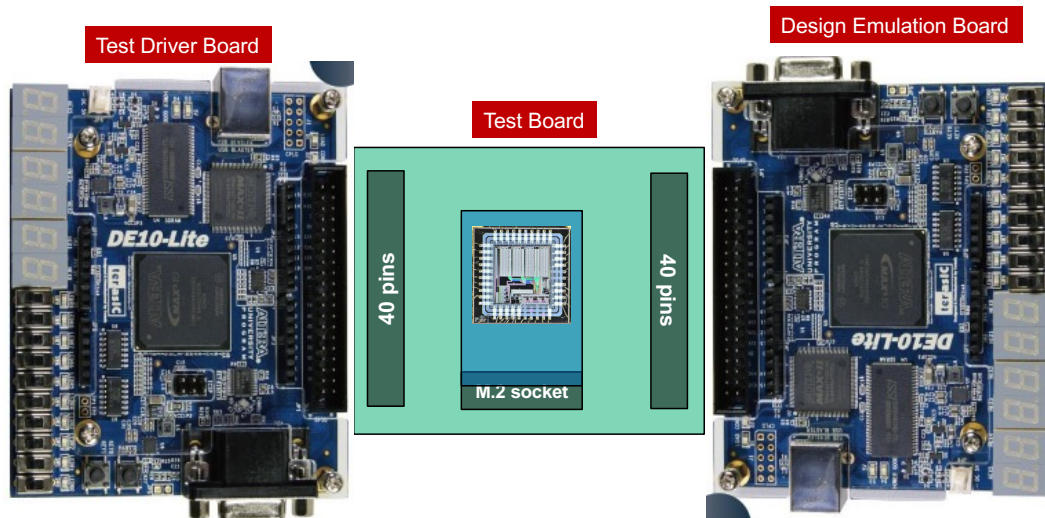
- ◆ **Boundary scan is accessed through five pins**
 - TCK: test clock
 - TMS: test mode select
 - TDI: test data in
 - TDO: test data out
 - TRST*: test reset (optional)
- ◆ **Chips with internal scan chains can access the chains through boundary scan for unified test strategy.**

The Test Access Port has four or five single bit connections:

TCK clocks tests into and out of the chip
TMS controls test operations
TDI test data into the chip
TDO (the only output) test data out of the chip; driven only when TAP controller is shifting out test data
TRST optional active low signal to asynchronously reset the TAP controller if no power up reset signal is automatically generated by the chip.

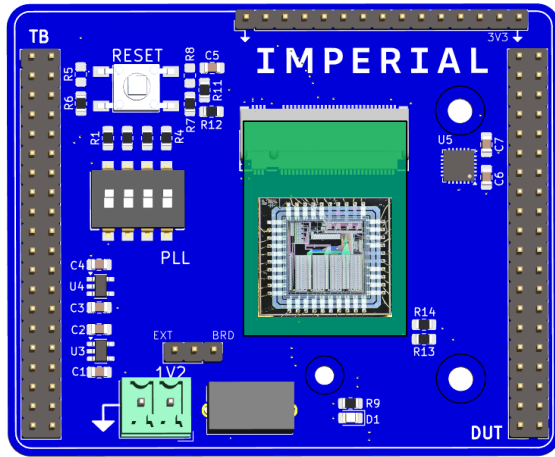
When the chip is in normal mode TRST and TCK are held low and TMS is held high to disable boundary scan. To prevent race conditions inputs are sampled on the rising edge of TCK and outputs toggle on the falling edge.

Test Environment

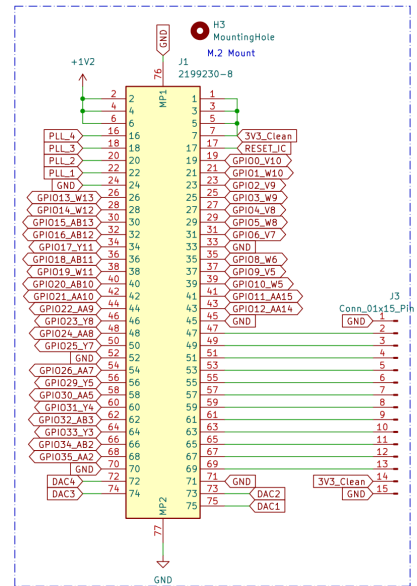


This is our test environment that we have described in my talk last week.

Test Board



M.2 Connector



This is M.2 connector for the daughter board. Details for the 40 pin connectors to the FPGA board is provided on the course webpage

Summary

- ◆ Think about testing from the beginning
 - Simulate as you go
 - Plan for test after fabrication

- ◆ “If you don’t test it, it won’t work! (Guaranteed)”